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Editor-in-Chief  
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Dear Editor,

I am pleased to submit my manuscript entitled **“PAR(2): A Phase-Gated Autoregressive Framework Reveals Tissue-Specific Circadian Gating of Cancer-Relevant Genes Across Mammalian Tissues”** for consideration in *PLOS ONE*.

This manuscript was previously considered at *PLOS Computational Biology* and is submitted here following transfer. The scientific content is unchanged. I believe *PLOS ONE* is well suited to this work because the journal’s scope encompasses methodologically rigorous, reproducible research across the biological sciences regardless of disciplinary boundary — and the PAR(2) framework sits at the intersection of computational biology, circadian medicine, and cancer biology in a way that makes cross-disciplinary placement appropriate.

### **What the paper does and why it matters**

I present PAR(2), a second-order autoregressive framework that extracts a single scalar quantity — the eigenvalue modulus  $|\lambda|$  — from two-parameter fits to gene expression time series. This metric quantifies *temporal persistence*: the degree to which a gene’s expression state carries forward across circadian cycles. The core analytical step is a two-parameter ordinary least squares regression, making the method computationally deterministic, parameter-light (three parameters total), and free from the convergence failures that affect maximum-likelihood state-space alternatives.

Applied to 3,588 clock-target gene interactions across 12 mouse tissues (GSE54650, Hughes Circadian Atlas) with replication in the gold-standard GSE11923 dataset (48 hourly timepoints), four principal findings emerge:

1. **Wee1 as the broadly conserved circadian gatekeeper.** The  $\text{Cry1} \rightarrow \text{Wee1}$  gating relationship is conserved across  $\geq 6$  tissues, making Wee1 the leading broadly conserved circadian checkpoint among the 23 cancer-relevant genes tested. Myc — often regarded as the canonical circadian-cancer target — is gated in only 2 tissues. Wee1 is additionally the gene whose eigenvalue sits closest to  $1/\varphi \approx 0.618$  across all 12 tissues ( $|\lambda| = 0.612$ ), providing a temporal analogue to Boman’s spatial Fibonacci observations in colonic crypt geometry.
2. **Nine-category persistence hierarchy.** Extension to  $\sim 1,594$  genes across nine functional categories reveals a fine-grained hierarchy: Clock > Chromatin > Metabolic > Housekeeping > Immune > Signalling > DNA Repair > Target > Stem Cell ( $H = 144.8$ ,  $p < 0.001$ ). Chromatin remodelling genes unexpectedly outrank housekeeping genes. Stem cell markers show the lowest persistence across all categories, reframing stemness as a low-memory dynamical state rather than a high-stability one.

3. **Root-space functional geography.** Plotting genes in AR(2) coefficient space reveals that functional categories occupy distinct dynamical neighbourhoods in the stationarity triangle. This “root-space geography” captures information orthogonal to expression-level PCA, demonstrated empirically side-by-side. A data-driven dynamical exclusion zone contains <3% of genes genome-wide.
4. **Drug target chronotherapy overlay.** Mapping 326 FDA-approved and investigational drug targets onto root-space identifies clock-gated targets amenable to tissue-specific time-of-day dosing optimisation, with 168 genes across 16 drug classes positioned relative to the four dynamical poles.

## Methodological rigour and reproducibility

The framework is validated against two mechanistic ODE systems (the Leloup–Goldbeter circadian clock model and the Boman colonic crypt compartment model), tested with four permutation null models (time-shuffle, expression-matched, label-shuffle, circular-shift), and subjected to an eleven-analysis robustness suite including leave-one-tissue-out cross-validation (hierarchy confirmed in 12/12 tissues independently), bootstrap resampling, population-level cross-validation (25/25 folds), and decomposition method sensitivity. Literature cross-validation recovers 58 of 59 curated circadian genes (98.3%) at  $\sim 180\times$  enrichment over background. The framework has additionally been replicated across 22 GEO datasets spanning six species (mouse, human, baboon, Arabidopsis, Drosophila, yeast).

I report limitations explicitly: PAR(2) is a descriptive discovery framework (cross-validation win rate 45.2%), not a predictive model. Moderate single-tissue FDR ( $\sim 16\%$ ) is reduced to  $\sim 1\text{--}5\%$  by cross-tissue consensus filtering. The framework operates on bulk-tissue data and cannot resolve cell-type heterogeneity within a tissue.

## Transparency and open data

All source data are publicly available from NCBI GEO. Analysis code, datasets, and pre-computed results are available at <https://github.com/mickwh2764/PAR-2--Final-09-12-2025> and Zenodo (DOI: 10.5281/zenodo.14751750). An interactive Discovery Engine platform is accessible at <https://par2-discovery-engine.replit.app>. A preprint is deposited on Research Square (DOI: 10.21203/rs.3.rs-9283100/v1).

Large language models (Claude, GPT-4) and the Replit development platform were used for code development, visualisation, and manuscript drafting. All analytical decisions, framework conceptualisation, and falsification test design are the author’s own.

## Suggested reviewers

1. Dr John Hogenesch (Cincinnati Children’s Hospital) — circadian genomics and tissue atlas methodology
2. Dr Achim Kramer (Charité – Universitätsmedizin Berlin) — circadian chronotherapy and drug timing
3. Dr David Suter (EPFL) — gene expression memory and temporal persistence at single-cell resolution

This manuscript is not under consideration at any other journal. I confirm that the work is original,

the data are publicly available, and the methods are fully described and reproducible.

Sincerely,

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